

# CHAPTER – I

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## ➤ INTRODUCTION OF JAVA:

Java is a truly Object Oriented Programming Language . Java was created by James Gosling and his team SunMicrosystem in 1991. Java's old name was Oak and later renamed as Java in 1995(Oracle). Java is a reusable , Secure and Platform Independent . Java have core principle "Write Once, Run Anywhere".

## ➤ OVERVIEW OF PROGRAMMING LANGUAGE:

Programming language is a communication between human and Machine.

## ➤ TYPES OF PROGRAMMING LANGUAGE:

- Low Level Language - Assembly Code ( Mnemonic code) , Chips.
- Medium Level Language - C,C++.
- High Level Language – Java , Python.

### ➤ **HISTORY OF JAVA :**

- 1991- Java created by James Gosling . language is called Oak.
- 1995 – later rename as Java.
- 2014- every six month cycle to update the version.
- Latest version of Java is 26.
- Java include JDK,JRE and JVM.

### ➤ **WHY WE LEARN JAVA:**

Java is a robust , Secure and Platform Independent. And we use java in many apps like android, Web Application , Finance and Banking, etc,.....

### ➤ **COMPILER AND INTERPRETER:**

COMPILER: Compiler is to find a syntax error and source code (.java file) to convert bytecode(.class file) .

INTERPRETER: JVM is use the principle” WRITE ONCE RUN ANYWHERE” . Interpreter use JIT for speed and its convert byte code into machine code. Its auto check error in line by line .

### ➤ **JVM(JAVA VIRTUAL MACHINE):**

- JVM is a important to run Java Programs.
- JVM convert bytecode into machine code

JVM Does(work):

- Loads Code
- Verifies Code
- Execute Code – converts Bytecode into Machine Code using:
  - ✓ Interpreter
  - ✓ JIT(Just In Time compiler)
- Manage Memory

### ➤ **JRE(JAVA RUNTIME ENVIRONMENT):**

- It Include JVM & Libraries(pre – built code).
- JVM is using Javac compiler to compile the program
- Inside JRE:

- ✓ JVM
- ✓ Core Libraries( java.util, java.io,java.lang)
- ✓ Supporting files
- Example: Run Java program but not write or compile programs

#### ➤ **JDK(JAVA DEVELOPMENT KIT)**

- JDK include Write, compile, debug , and run java programs
- Its Inside :

- ✓ JRE

- ✓ JVM+ Core Libraries

- Development Tools:

- ✓ Compiler (javac)

- ✓ Debugger (jdb)

- ✓ Documentation tool (Javadoc)

#### ➤ **INTEGRATED DEVELOPMENT ENVIRONMENT:**

- Popular IDEs for Java:

- ❖ Eclipse

- ❖ IntelliJ IDEA

- ❖ NetBeans

- ❖ VS Code

- Programmer's Workspace.
- Complete environment to write , edit, compile, debug, and run code easily.

#### ➤ **MAIN FEATURES:**

1. Code Editor
2. Compile/Interpreter Integration
3. Debugger
4. Project Management
5. Other Tools ( build tools, version control(Git), Plugin Support, Testing frameworks)

➤ **JAVA'S MAGIC BYTECODE:**

- If compile a program , becomes a .class file (bytecode)
- Every .class file starts with special code 0xCAFEBAFE.
- Secret Signature start of every java .class file. This is called Magic number.
- Its like a ID card of bytecode.
- When JVM opens a .class file first thing it checks whether it starts with special code.
- If its there – JVM Knows it's a valid Java bytecode files.
- If not – JVM rejects it (Prevents error).